

Name:

Student ID:

Agentic AI for Business and FinTech (SEEM5660)
Individual Homework 01, Due date: 29 September, midnight.

Individual homework. You are encouraged to discuss with others; however, you must submit your own work.

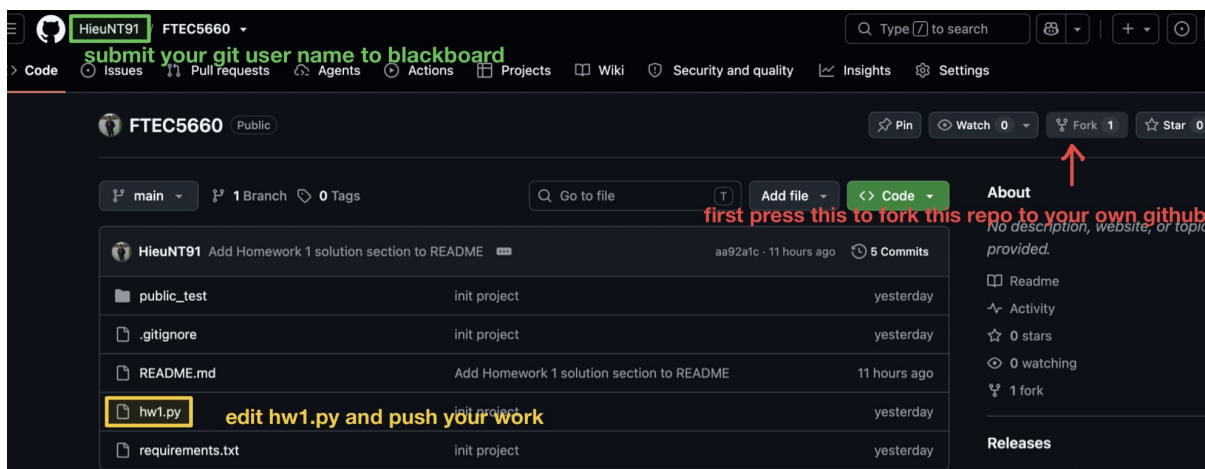
Task 1: Implementation (90 points)

- **Data.** `public_test/` in the template repo has 7 sample receipts plus `ground_truth.json` (the correct answers), so you can check your own accuracy. We will evaluate your implementation on a private folder of unseen receipts.
- **Setup:** Fork <https://github.com/HieuNT91/FTEC5660>, then:

```
git clone <your fork's url>
cd FTEC5660
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
echo "DEEPSEEK_API_KEY=your_key_here" > .env
```

Never commit `.env` by ensuring `.env` is always in `.gitignore`.

- **Implementation details:** You are required to use `deepseek-v4-flash-vision-exp` as the backbone model and `LangChain` to implement your chain.
- **Submit:** a GitHub repo named **FTEC5660** containing `hw1.py`, `requirements.txt`, and `README.md`. Submit your GitHub username on Blackboard.
- For students unfamiliar with GitHub, we recommend the following introductory tutorial: <https://www.youtube.com/watch?v=-RZ03WHqkaY>.
- If you encounter any problems with this homework, please inform the TA immediately. The TAs responsible for this homework are Binh Nguyen (tbnguyen@se.cuhk.edu.hk), WeChat ID: `binhnguyen2702` and Hieu Nguyen ([1155203613@link.cuhk.edu.hk](https://t.me/link.cuhk.edu.hk), WeChat ID: `junhill19961`).



1 Descriptions

Given a folder of supermarket receipt images, design a chain to answer two fixed questions about the whole folder (one number each, summed across every receipt):

1. *How much money did I spend in total for these bills?* The final payment (after **ROUNDING**) on each receipt, summed.
2. *How much would I have had to pay without the discount?* Each receipt's **SUBTOTAL** plus every discount/promotion/coupon line added back as a positive number (do *not* add back **ROUNDING**), summed.

Example (receipt5.jpg): items \$10.00 + \$36.90 + \$60.80, then 5% OFF -\$5.39, SUBTOTAL \$102.31, ROUNDING -\$0.01, OCTOPUS \$102.30. Query 1 answer = 102.30. Query 2 answer = 107.70 (102.31 + 5.39), not 102.31 or 102.30.

2 Your Implementation

Fill in exactly two functions in `hw1.py`, both marked `### YOUR CODE HERE`:

- `build_chain()` set up your model (`deepseek-v4-flash-vision-exp`) and prompt(s) once.
- `answer_queries(chain, images)` return a dict keyed by the two exact question strings above, each value a response containing **one** HKD amount, e.g. "HK\$1974.30".

Do not touch anything else in the file; the rest (argument parsing, CSV writing) is provided, already works, and there are nothing protecting it, so leave it alone to keep it working.

Submission Requirements

Your submission MUST run end-to-end and produce results.csv.

A submission that crashes, hangs, or produces no `results.csv` scores **zero** on the automated part, no matter how good the chain design is. Test from a fresh clone before you submit. Make sure your GitHub repository is public and accessible at submission and during grading.

```
python3 hw1.py --image-folder public_test
cat results.csv
```

This writes `results.csv` with the columns `query`, `model_response`, `correctness` — checked automatically against `ground_truth.json` when present. Each response must contain exactly one number: no stray extra digits (percentages, item counts, a second candidate amount), or it may be marked incorrect even if the right value is in there.

Complete your report in the “Homework 1 solution” section of `README.md`. Include a visualization of your chain design and one paragraph describing your solution.

3 Evaluation

At test time, we will evaluate your implementation on a private set of receipts across three independent runs. You are encouraged to test your chain thoroughly on different combinations of receipts from the public set and aim for the highest possible accuracy. Your final evaluation will be based on your implementation's performance across the private test runs.

Task 2: Reflection (10 points)

Submit your reflection about how the AI-related events that happened in the past 10 days have changed your perspectives/career plan. No restrictions on the format; you can write/draw/or submit a AI-generated submission.